ESP DOMAIN (earthed via USB/PC allowed)

ISOLATED SDC DOMAIN (referenced to SDC X5 only)

PS1 1S4E_0509S1U: unregulated 9 V (~9–11 V at our load), needs >= 10 mA load:
R24 + U4 quiescent current + VDD_ISO load. U4 L78L05 makes VDD_ISO = 5 V.
Pins 2 (+Vin) and 3 (−Vout) straddle the isolation gap: narrow oval pads.

AD5272: ADDR = GND -> I2C 0x2F. RESET tied to VDD.
Firmware: write control reg C1=1 at boot, never set C0 (50-TP).

SDC service port (J5): UART at 5 V logic, isolated by U3. Pinout of the SDC socket
is not yet known: J5-2 RX must go to the SDC's TX, J5-3 TX to the SDC's RX.
J5-1 is GND_ISO (= SDC X5 ground): only connect if the port GND is the same potential.
R17 (OR): remove if the SDC port has its own 5 V supply pin.

DevKit: ESP32-S3-DevKitC-1 (N16R8 clone), header rows 25.1 mm apart (measured).
GPIO4 = RELAY_EN (ESPHome restore_mode ALWAYS_OFF), GPIO5 = RELAY_SENSE (LOW = coil on),
GPIO8 = SDA, GPIO9 = SCL, GPIO6 = service RX, GPIO7 = service TX (UART),
GPIO1/GPIO2 = expansion header J6.

Relay K1: coil 1/16, COM 4/13, NC 6/11, NO 8/9 (Omron datasheet).
Both poles in parallel for dry-circuit reliability.
De-energised: VE1 -> R10 (fail-safe). Energised: VE1 -> R9 -> AD5272 W-A.

Generated by tools/make_schematic.py from tools/design.py – edit design.py, not this file
ESP32-S3 + ISO1540 + AD5272 + fail-safe relay, see docs/design-notes.md

Sheet: /
File: sdc-ntc-emulator.kicad_sch

Title: SDC NTC emulator – wireless room sensor for Smile SDC3-40PM

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